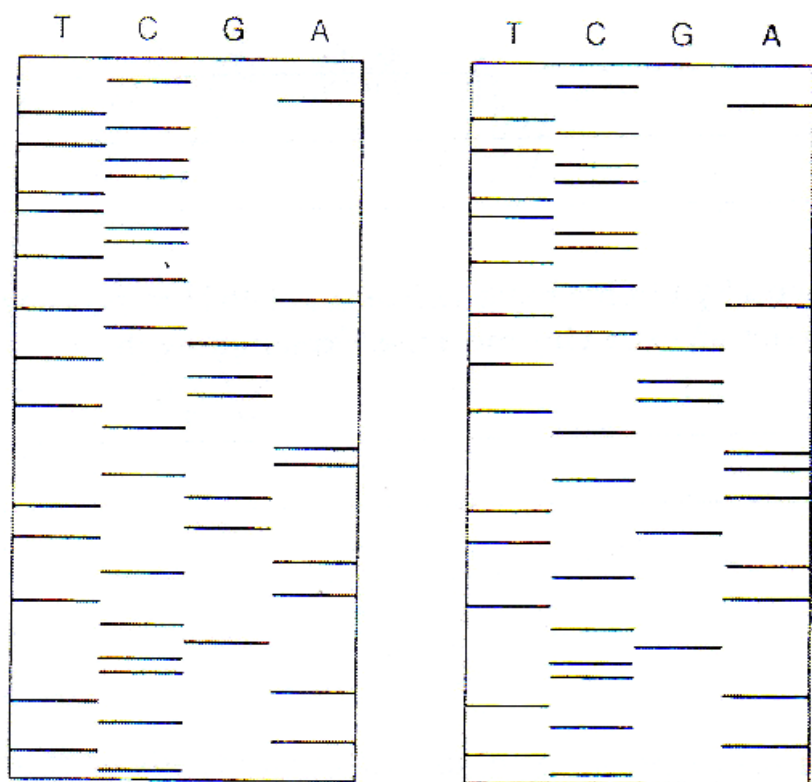


Section A: Structured questions

Question 1 [Total: 12m]



Part of the mutant allele

Part of the normal allele

(a) From the sequence in above figure, identify

- (i) the base that is missing in the abnormal sequence; [1]

Adenine [1] A [0.5]

reject adenosine

- (ii) the base that has been put in its space [1]

Guanine [1] G [0.5]

reject guanosine

- (iii) Name the type of mutation that occurred [1]

Substitution/point mutation [1]

(b) Describe how a single base in the gene leads to the structural and functional changes of the ethylene forming enzyme (EFE). [4]

mRNA sequence/ codon sequence will be changed [0.5] in transcription [0.5].

seq of amino acids encoded will be different [0.5] in translation[0.5].

R groups of the amino acid will be different [0.5].

Interactions [0.5] will be different – folding of protein [0.5].

3 d config will be different [0.5].

Active site configuration will be changed [0.5].

(c) Using the information above, suggest why mutations are recessive. [1]

If a non functional protein formed by an allele, the other allele can produce the enzyme that can mask the non-functional protein. Thus if a non-functional protein is formed, there is no expression of that allele, that allele is considered to be recessive.

d) Suggest why single base substitutions do not always have such a major effect on protein structure. [2]

DNA code is degenerate; [0.5]

more than one codon [0.5] can code for the same amino acid. [0.5]

especially if the substitution occurs on the last base in the codon [0.5].

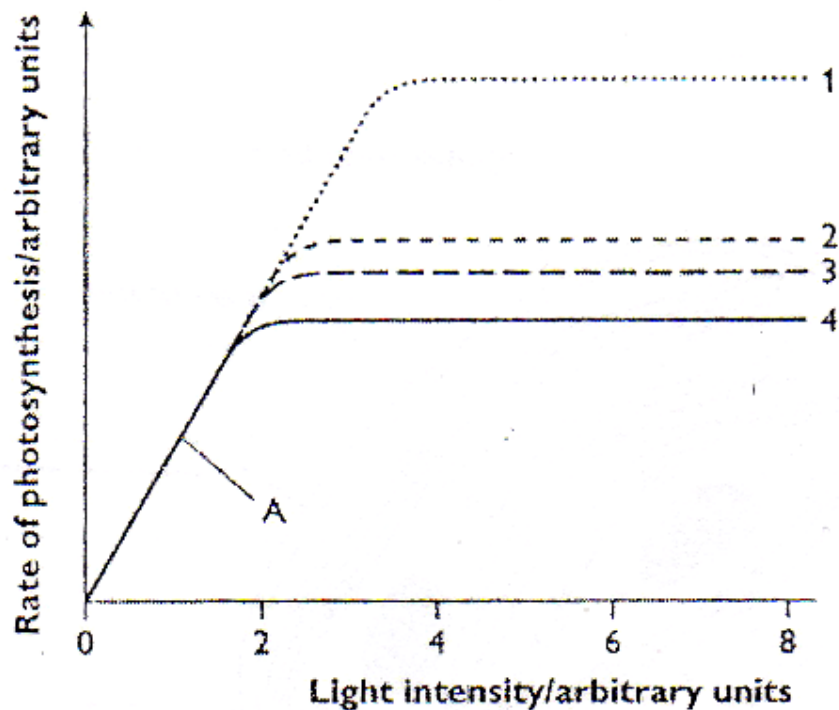
e) Define chromosomal mutation using a named example. [2]

Chromosomal mutation is caused by changes to the number or structure of chromosomes [1].

Change in number of chromosome [0.5] aneuploidy [0.5] – trisomy 21[0.5], Down syndrome [0.5]; trisomy

Change in structure of chromosome [0.5] – translocation [0.5], fragile X syndrome [0.5]

Question 2 [Total: 12m]



Line	CO ₂ concentration / %	Temp / °C
1	0.06	30
2	0.06	20
3	0.03	30
4	0.03	20

Fig. 2.1

- (i) State which factor is most likely to be limiting the rate of photosynthesis at point A. [1]

Light intensity [1],

Reason: The rate of photosynthesis increases together with an increase in light intensity.

(ii) The growth of crops in glasshouses in winter is limited by the rate of photosynthesis. A glasshouse has a light intensity of 4 arbitrary units, a temperature of 20°C and a carbon dioxide concentration of 0.03%.

Which one of the following conditions, X, Y or Z, would result in the greatest increase in growth of the crop plants?

X - increasing the carbon dioxide concentration from 0.03% to 0.06%

Y - increasing the temperature from 20°C to 30°C

Z - increasing the light intensity from 4 to 8 arbitrary units

Explain your answer. [2]

'X [1];

From 2 A.U., [0.5], light intensity is no longer a limiting factor, [0.5]

whereas increasing the CO₂ concentration will increase the rate of reaction [0.5].

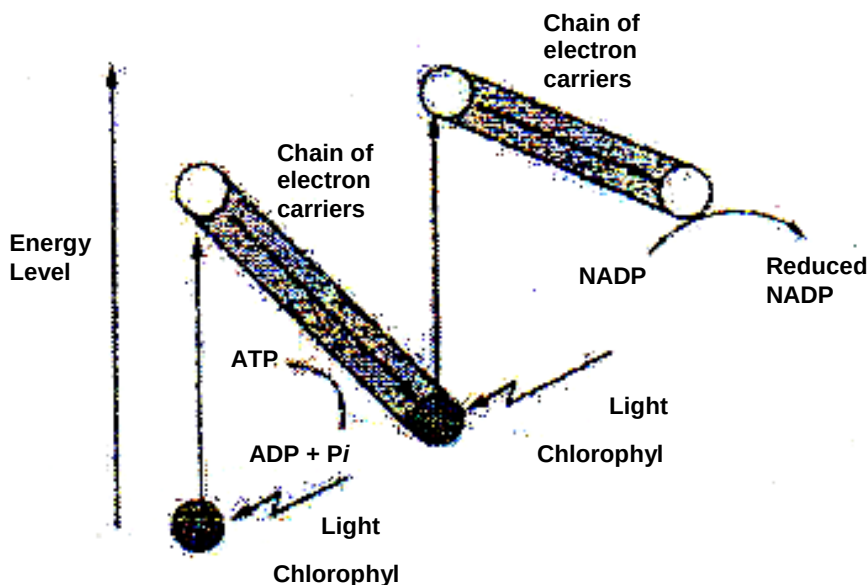
(iii) Compare and explain the effect of increasing carbon dioxide concentration at 20°C and 30°C. [2]

Comparing graphs 2 & 4, at 20°C, the rate of reaction increase to a smaller extent as compared to graphs 1 & 3, where at 30°C, the rate of reaction increase to a greater extent [1]

– A.W. – refers to graph [0.5], mentions the difference in rates of reaction [0.5].

@ Higher temperatures, enzymes and substrates have higher kinetic energy [0.5]; more effective collisions [0.5]; higher rate of reaction [0.5].

ATP is produced in the light-dependent reaction of photosynthesis. The diagram below shows the energy changes which take place in the light-dependent reaction.



The weedkiller DCMU blocks the flow of electrons along the chains of electron carriers. Describe and explain the effect this will have on the production of triose phosphate in the light-independent reaction. [3]

Without the flow of electrons, electron transport chain (ETC) will stop functioning [0.5];

No / low production of NADPH [0.5], ATP [0.5] – no reducing power [0.5] & ATP for the Calvin cycle [0.5].

Thus glycerate phosphate [0.5] is reduced [0.5] to triose phosphate;

3-phosphoglycerate utilizes ATP to give 1,3 biphosphoglycerate [0.5];

1, 3 biphosphoglycerate gets reduced by NADPH to triose phosphate/ glycerate-3-phosphate [0.5].

(c) Cellulose is the major component of cell walls in plants. Fig 2.2 shows part of a cellulose molecule

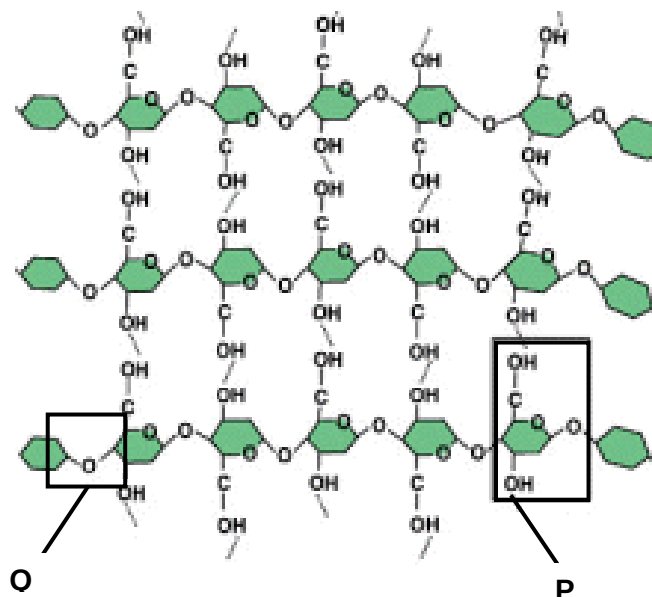


Fig. 2.2

(c)(i) Name the molecule labelled P. [1]

β -glucose [1]

reject glucose

(c)(ii) Name the bond labelled Q and outline how it is formed. [1]

Bond formed: β - (1, 4) glycosidic bond [0.5],

Process: Condensation / polymerisation [0.5]

(c)(iii) Using the information from the diagram, explain why cellulose is suitable as material for the plant cell wall. [2]

Strength – straight chain polymer; multiple hydroxyl groups [0.5] on the glucose residues from one chain form hydrogen bonds [0.5]; with oxygen molecules on the same or on a neighbor[0.5] chain, holding the chains firmly together side-by-side and forming microfibrils, followed by macrofibrils with high tensile strength[0.5].

The overall structure is a mesh of bundles of fibrils. Within this network are extensive pores which allow movement of water and minerals.

Question 3 [Total: 12]

(a) State what is meant by “dual control of the lac operon” [2]

The lac operon is under dual control:

- negative control by the lac repressor
- Positive control by cAMP receptor protein.

Common mistakes:

(1) Control by presence of glucose and lactose

(2) Confusion of terms: inducible & repressible (often used interchangeably)

(b) With reference to Fig.1, explain how expression levels of β -galactosidase is regulated by the **lac repressor** from time X to Y. [6]

X:

- Both glucose and lactose are present [1/2] and glucose is used but not lactose [1/2]
- In the presence of lactose, allolactose binds to lac repressor. [1/2] Thus, the Lac repressor is in its inactive configuration and does not bind to the operator [1/2]
- But another mechanism prevents transcription of the lac operon due to catabolite repression [1/2] (i.e. Catabolite regulator protein (CRP) is inactive and disengages from CRP binding site)
- RNA polymerase is unable to bind to the promoter [1/2] / Low β -galactosidase mRNA is produced [1/2]

[max 3m]

Common mistake: allolactose binds to the active site of repressor

Y:

- Once the glucose concentration has depleted, *E. coli* uses lactose. [1/2]
- β -galactosidase present in the bacterial cells will catalyse conversion of some lactose molecules to allolactose [1/2]
- Allolactose binds to the lac repressor [1/2] and changes configuration / conformation of lac repressor [1/2]
- Lac repressor is inactive and does not bind to the operator [1/2]
- RNA polymerase is able to bind to the promoter to transcribe the mRNA of the lac operon / Lac operon is switched on / High β -galactosidase mRNA is produced [1/2]

[max 3m]

(c) Describe another mechanism which regulates the expression of β -galactosidase at Y.
[2]

- In low levels of glucose, cAMP levels increases, CRP is active and it binds to the CRP binding site.
- Transcription of B-gal proceeds at a high level.

Common mistake: cAMP binds to CAP binding site directly

(d)(i) Suggest one mechanism that causes the amount of β -galactosidase to decrease.
[1]

- mRNA degradation / protein is degraded

Common error:

– end product inhibition

(ii) The bacterial cells were found to be viable. Suggest how the bacteria can survive without a carbon source. [1]

- Stress genes can be turned on which allows the bacterial cells to survive under stress or starvation

Knowledge on control and organisation of prokaryotic & eukaryotic genomes required here

Reject:

- (1) *E.coli* cells photosynthesise to make own sugars
- (2) survives through anaerobic respiration

Question 4 [Total: 8]

(a) With reference to Fig. 4.2, state how the levels of ras and p53 proteins correlate with the development of cancer. [2]

- In the cancerous tissue, there are elevated / higher levels of ras protein as compared to the normal tissue. [1]
- Also, the p53 protein level is diminished / lower in cancer tissue than in the normal tissue [1].

To obtain full marks, there must be comparisons made between level of BOTH proteins in both normal and cancerous tissue.

(b) State the role of the *ras* gene in relation to the development of cancer. [2]

- *Ras* gene is a proto-oncogene. [1/2]
- High levels of expression ras protein [1/2] constantly stimulates cell division [1/2] ,
- Hence leading to formation of cancerous tissue / tumor [1/2].

(c) Explain how mutations in the *p53* gene may lead to uncontrolled cell division. [4]

- *p53* is a tumor suppressor gene [1/2] involved in the inhibition of cell division [1/2].
- Mutation in both alleles of the *p53* gene [1] leads to a loss of function mutation [1].
- When there is DNA damage, the mutated *p53* protein is unable to:
 - repair the DNA damage [1/2]
 - stimulate the cell to undergo apoptosis [1/2]

Common errors:

- loose use of the terms “gene”

Question 5 [Total: 13]

(a) Describe the interaction of the alleles,

- (i) at the locus **G/g**, controlling green-based or not green-based fruit; [1]
 - green-based is dominant to uniform colour / **G** is dominant to **g**;
- (ii) at the locus **R/r**, controlling red or orange fruit. [1]
 - red fruit is dominant to orange fruit / **R** is dominant to **r**;

Common error: a number of students suggest that the conditions were due to epistasis.

(b) Using the symbols given in (a), state the genotype of variety **B**. [1]

- **ggrr / rrgg;**

(c) Plants from the F₁ generation were test crossed (backcrossed) to variety **B**. The ratio of phenotypes expected in a dihybrid test cross such as this is 1:1:1:1.

Using the symbols given in (a), draw a genetic diagram of the test cross to show that the expected ratio of offspring phenotypes is 1:1:1:1. [4]

F1 generation phenotypes: green-based fruit X uniform orange
F1 generation genotypes: GgRr ggrr

Gametes:



1m

Offspring genotypes	GgRr	Ggrr	ggRr	ggrr	1m
Offspring phenotypes	Green-based red	Green-based orange	Uniform red	Uniform orange	1m
Offspring phenotypic ration	1	1	1	1	1m

The correct genotype must be mentioned together with the correct corresponding phenotype to obtain full marks.

(d) (i) State the number of degrees of freedom applicable to these data. [1]

- **3**

(ii) Use the calculated value of χ^2 and the table of probabilities provided in Table 5.1 to find the probability of the results of the test cross departing significantly by chance from the expected ratio. [1]

- **> 0.1**

(iii) State what conclusions may be drawn from the probability found in **(d) (ii)**. [2]

- **Difference from expected results is not significant / statistically different;**
- **The calculated χ^2 value of 3.2 is less than the critical value of 7.82 at p = 0.05; the null hypothesis should be accepted;**
- **Results are due to chance;**
- **The two gene loci are not linked / alleles at the two gene loci assort independently;**

½ m per point

(e) Experiments have shown that loci **G/g** and **R/r** are on the same chromosome of the tomato plant genome. The two loci are 44 map units apart.

Explain how the results of the test cross shown in **(d)** could occur when the two loci are on the same chromosome. [2]

- **The two gene loci are too far apart (for linkage to be detected); [1/2]**
- **Correct reference to recombinants; [1/2]**
- **Crossing over has occurred; [1/2]**
- **Details of crossing over:**
 - **[occurs in prophase I of meiosis / exchange of (part of) non-sister chromatids]; [1/2]**
***ignore chiasmata*;**
- **Crossing over has occurred twice / even number of times; [1/2]**
- **(therefore) restoring loci to parental combinations; [1/2]**

Common error: some reference to the two gene loci located on different chromosomes / reference to independent assortment when (e) stated clearly that the two loci are on the same chromosome and the map distance between them is 44 map units apart – i.e. well-separated.

Question 6 [Total: 13]

(a) Explain why the blood glucose concentration of individual A did not return immediately to 90 – 100 mg per 100 cm³ blood. [2]

- Time delay in blood glucose concentrations returning to the norm of (90-100 mg/100cm³ of blood) [1/2] OR
- It takes 120 minutes from the start of the experiment for blood glucose concentration of individual A to return back to the initial blood glucose concentration of ~ 90 mg/100cm³ of blood [1/2]

Linked to

- Reason why insulin takes time to work [1] ;
 - e.g. slow release of insulin from beta cells of the islets of Langerhans in the pancreas;
 - Time required for liver cells to absorb glucose etc.

Reject: insulin released from α -cells of islets of Langerhans

- Link points 1 & 2 to homeostasis; [1/2]
- Leading to fluctuating levels of blood glucose around a set point / the norm (90-100 mg/100cm³ of blood); [1/2]
- dynamic equilibrium; [1/2]

(b) Explain the data for individual B given in Table 6.1. [3]

- Blood glucose concentration of Individual B rises from 100 mg / 100cm³ of blood at 0 min to a maximum of 208 mg / 100cm³ of blood at 60 min, and remained above the norm (195 mg / 100cm³ of blood) even after 120 min; [1/2]
- Individual B has diabetes (mellitus); [1]
- glucose not converted to glycogen for storage; [1/2]
- or glucose not absorbed into, cells / liver / muscle ; [1/2]
- therefore blood sugar rises / stays high ; [1/2] OR minimal drop in blood glucose concentration from 208 mg / 100cm³ of blood at 60 min, to 195 mg / 100cm³ of blood) at 120 min; [1/2]
- until glucose used as substrate for respiration / excreted in urine ; [1/2]
- AVP ; e.g. description of Type 2 diabetes (max 1)

(c) (i) Individuals with Type 2 diabetes are recommended to make the following changes to their diet:

- avoid foods high in refined sugar (a high glycaemic index)
- eat carbohydrate in the form of more starch (a low glycaemic index)
- reduce the intake of saturated fat.

Explain why it is important to avoid foods high in refined sugar but to eat more starch. [2]

- **refined sugar is soluble / easily digested ; [1/2]**
- **easily / quickly, absorbed ; [1/2]**
- **(therefore) surge of glucose in blood / rapid increase in blood glucose concentration / AW ; [1/2]**
- **starch is a complex carbohydrate / polysaccharide ; [1/2]**
- **slowly release into blood ; available as required / AW ; [1/2]**

Common error:

- **repeating question in the answers**

(ii) Suggest why diabetics should eat less saturated fat. [2]

- **excess glucose in diabetics is converted to and stored as fat / AW ; [1/2]**
- **lipogenesis ; [1/2]**

Points 1 & 2 must be discussed together

- **saturated fat carried with cholesterol / would raises (blood) cholesterol levels ; [1/2]**
- **deposited in walls of arteries ; [1/2]**
- **obstructs lumen of coronary artery / leads to narrowing of lumen of coronary artery ; [1/2]**
- **causes coronary heart disease / heart attacks / myocardial infarction; [1/2]**
- **formation of plaques / atheroma / atherosclerosis [1/2]**

Common error: fat converted into sugar and this has an effect on blood glucose level

(d) Using the information in Fig. 6.1, explain why diabetics are encouraged to use a combination of diet and statin, rather than diet or statin alone, to treat their condition. [4]

- LDL lowered most (mean change of -40%) by diet and statin combined; Accept cholesterol for LDL
- LDL also lowered by, diet (mean change of -11%) / statin (mean change of -30%) alone;
- effect on LDL cumulative / additive / AW ;
- Insulin concentration / insulin resistance, increased by statin only (mean change of +13%);
- Insulin concentration / insulin resistance, decreased by diet only (mean change of -14%);
- insulin concentration / insulin resistance, diet and statin combined
- Balanced each other out / minimal change (mean change of -1%);
- comparative figs in support ;
- AVP ; A ref to groups

To obtain full marks:

Discussion should include

- LDL cholesterol
- Insulin
- Insulin resistance

Question 7

a) Using your understanding of Darwin's theory of evolution and information from the diagram, suggest why the fossils have many similar characteristics to the smaller species in the **same region**. [4] comprehension

- Descended from a common ancestor [0.5];
- Same region – similar environmental factors [0.5];
- similar selection pressures [0.5];
- features that are selected for are the same/convergent evolution / homologous structures [0.5];
- adapted to the environment [0.5];
- thus organisms with this set of features will have a selective advantage [0.5] and enjoy reproductive success [0.5];
- thus passing down the genes to their offspring resulting in the features being passed down in the populations [0.5];

- b) Using the information from the diagram and the passage, explain why the smaller armadillo is still living whereas the giant armadillo is extinct. [2] comprehension

Choose any of the possible answers below:

- **Loss of teeth [0.5].** – No need to waste resources on teeth as they are not required for the sucking up of insects [0.5]; the loss of teeth is a selective advantage [0.5] as compared to the toothed armadillo; gene passed down to offspring, leading to gene coding for the loss in teeth increasing in the population[0.5].
- **Segmented carapace** can fold better into a complete armour [0.5] for protection against predators [0.5]; as compared to a fused carapace. Having segmented carapace is selectively advantageous [0.5] as compared to the fossil armadillo; gene passed down to offspring, leading to gene coding for segmented carapace increasing in the population [0.5].
- **Bigger claws [0.5]** for the modern species thus they can dig into the termite mounds more effectively to obtain their food source [0.5] as compared to extinct species. Having bigger claws is selectively advantageous [0.5] as compared to the fossil armadillo. Gene for bigger claws passed down to offspring, leading to gene frequency increasing in the population [0.5].

Of the 20 species of living armadillos, 12 are listed as Vulnerable, Endangered, Near Threatened, or Data Deficient. It is believed that the number of deaths of the armadillos increased drastically in areas densely populated with human beings.

- c) Using the information from the passage and your understanding of the human environment, suggest why the adaptations of small armadillos are not endowing them with selective advantage in areas with high human populations. [2]

Choose any of the possible answers below:

The evolved trait of folding up into an armored ball [0.5] will not be able to defend against vehicles [0.5], thus they are selected against even if they evolved the trait. [1]

Pets of humans / humans [0.5] hunting the armadillos; not predators that the armadillos has evolved defence for [0.5] Thus they are selected against even if they evolved the trait. [1]

Loss of habitats / no loose soil /loss of ants and termites [0.5], the armadillos have traits evolved for hunting insects [0.5] Thus they are selected against even if they evolved the trait for hunting insects. [1]

*****reject decrease in no of insects because not all insects decrease with an increase in human population. *****

- d) Explain how you can use other methods (other than comparing the anatomy) to determine the evolutionary relationship between the giant and living armadillos. [2]
- Comparing DNA sequences / amino acids sequences obtained from the organism; [0.5].
 - DNA hybridisation/ sequencing [0.5].
 - Level of similarity [0.5].
 - Higher similarity – more related [0.5].

Section B: Essays

Question 8

- (a) Describe the general lifecycle of a retrovirus using a named example. [8]

One named example will be the **Human immunodeficiency virus (HIV)**

- o The **adsorption stage** involves the binding of **gp 120 receptors** on the HIV virus to **CD4 molecules and chemokine receptors** found on the surface of human **T4-lymphocytes and macrophages**.
- o In the **penetration stage**, the virus enters the cell by **fusing** with the host cell membrane.
- o Next, in the **uncoating stage**, the viral capsid is digested by **enzymes** in the cell.
- o The RNA genome and the enzyme reverse transcriptase are released.
- o Under the action of **reverse transcriptase**, the **viral RNA** acts as a **template for making a single strand of complementary DNA (cDNA)**. The reverse transcriptase then **synthesizes another DNA strand using the single stranded cDNA as a template**.
- o The resulting double stranded DNA **inserts itself** / integrates into the chromosomal DNA of the host cell, forming a provirus.
- o The provirus either **lies dormant** or goes into the **maturation stage**
- o In the maturation stage, mRNA is manufactured from the provirus that is then **translated into proteins**. These proteins e.g. for **new viral coat and envelope proteins** are required for assembly of new viral particles.
- o **RNA genome** will also be encoded in the maturation stage. The newly synthesized RNA genome will travel to **the cell cytoplasm**. The proteins of the **viral cell wall** will travel to the **host cell surface membrane**
- o The viral membrane proteins are gathered in the membrane in the virus coat region;

- o initiating the budding of this membrane region and enclosing the virus within a membrane envelope. The virus particles exit the host cell through endocytosis

(b) Compare the general structures of a bacteriophage and an animal virus. [6]

Similarities

Both kinds of viruses

- o have genomes that is made up of RNA or DNA
- o have capsid coats that are made up of capsomeres
- o have receptor molecules on the surface of the outermost coat
- o do not contain any organelles
- o do not have cytoplasm

Differences

Most animal viruses have an outer coat surrounding the nucleocapsid known as the envelope whereas the majority of bacteriophages do not have an envelope outside of their nucleocapsid.

(c) Describe how viruses can induce diseases in animal cells. [6]

Viruses can cause disease in animals by damaging host cells or by evading host immune system.

- Cytopathic effect damages infected host cells by (any 4 from below)
 1. directly lysing the host cell by causing cytolysis of the infected host cell
 2. Interrupting normal host cell DNA /RNA / protein synthesis. Virus can also create nicks or breaks in the host cell's chromosomes. Both will result in structural or functional defects in the infected host cell leading to cytolysis.
 3. The production of viral glycoproteins will result in the changing of the surface antigen of the host cell's cytoplasmic membrane resulting in the host cell being recognized as foreign and destroyed by the body's immune defenses such as the phagocytes.
 4. The virus will take over the control of cell functions and resources, depleting the host cell of cellular materials essential for normal function of the host cell;
 5. stimulating body cells to release inflammatory cytokines and chemokines or stimulates the body cells to release inflammatory vasoactive peptides to result in vasodilation and increased mucous secretion;

6. Inducing adjacent host cells to fuse together forming giant multinucleated cells (syncytia). Cell fusion causes loss of individual cellular integrity and will eventually lead to cell death.
7. Oncogenic viruses play a role in normal cells becoming malignant. Some viruses can infect a cell and turn on the gene for cell division when it should be off.

- Evading the host immune system (any 2 answers from below)

1) Antigenic shift that can lead to large changes in the viral membrane proteins or genome;

Antigenic drift, gradual changes made to the viral membrane proteins or genome;

A high rate of mutation that resulted in the virus changing rapidly that leads to the host cell not being able to recognize the viral particle.

2) Numerous viruses can block and alter the formation of MHC-I molecules by the infected cell to escape the natural killer (NK) cells.

3) Producing proteins that blocks apoptosis so that the virus can continue to infect more host cells.

Question 9

(a) Describe the control mechanisms involved in the expression of eukaryotic genes. [8]

Chromatin modification	▪ Plays a significant role in transcriptional regulation. ▪ Genes in the heterochromatin region, which is more highly condensed, are usually not expressed ▪ DNA methylation prevents gene expression when methyl groups are added to the DNA ▪ DNA methylation is essential for the long term inactivation of genes ▪ Histone acetylation increases the expression by adding acetyl groups to histones ▪ which affect how compacted the region is, therefore affect transcriptional activation
Regulatory sequences	2 types of control elements present: ▪ Basal / general transcription factors bind to proximal control elements which are required for transcription of all genes ▪ Specific transcription factors (activators) bind to distal control elements (enhancer) which act in a tissue-specific / time-specific manner. These are protein activators which bind to distal control elements associated with specific genes, and activate transcription of these specific genes. ▪ This aids in the folding of DNA which aids in RNA polymerase accessibility
Induction / repression in response to external stimuli	In response to external stimuli, transcription factors may bind to (or silencer) regions, thereby activating (or silencing) transcription
Post transcription	Alternative RNA splicing: Different mRNA molecules are produced from the same primary transcript Regulation of mRNA degradation ▪ enzymatic shortening of the poly (A) tail which triggers the action of enzymes to remove the 5' cap ▪ Once the cap is removed, nuclease enzymes chew up the mRNA from the 5' end

Translation	Control binding of ribosome/prevent ribosome from translating/block ribosome attachment/ initiator tRNA attachment by regulatory proteins at the leader regions at 5' end of the mRNA
Post translation	Activity of protein can be regulated by (a) end product inhibition (b) covalent modification /phosphorylation (c) Protein tagged with ubiquitin and recognised by proteasome will be degraded: regulates length of time protein functions in the cell.

Max 8m: to obtain full marks, student should have a good discussion of aspects of eukaryotic gene expression at all levels

(b) Describe the differences in structure between the prokaryotic and eukaryotic genome [6]

Feature	Prokaryotic genome	Eukaryotic genome
Ploidy	The genome is haploid	The genome is diploid
Number of chromosomes	There is only one chromosome.	There may be more than one chromosome.
Size & complexity	The genome is smaller and simpler.	The genome is larger and more complex
Nature of chromosome	The DNA is circular.	The DNA is linear
Presence of extrachromosomal DNA	Yes, in the form of plasmids, small circular DNA molecules which replicate independently from the main bacterial chromosomal DNA.	None within the nucleus, but mitochondrial and chloroplast DNA are found in mitochondria and chloroplasts respectively
Degree of condensation	Compaction process involves: (a) Formation of loop domains. The genome of the prokaryote is condensed and it is folded into 50 to 100 domains (b) DNA supercoiling	Condensation of DNA occurs at multiple levels with high degree of complexity : DNA wound around proteins to form nucleosome, which folds into the 30nm chromatin fiber (solenoid), which then compacts further into looped domains/metaphase chromosome which finally compacts into the chromosome structure
Association with proteins	Yes, but relatively lesser association; wound around nucleiod-associated proteins which facilitate the folding of the genome.	Yes, associated with large amounts of proteins, e.g. histones, scaffold proteins.
Regions of chromosome	The chromosome is circular DNA with no distinct specific regions	The genome has specific distinct regions such as the (a) centromere (b) telomere
Replication	There is a single origin of replication	There are multiple origins of replication
Location	Not enclosed by any membrane; found in a region of cell known as nucleiod.	Enclosed by a nuclear membrane; thus separated from the rest of the cytoplasm.

Max 6

(c) Explain the role of telomeres in maintaining the integrity of a eukaryotic organism. [6]

1. **Definition: Telomeres are non-coding DNA sequences found at the ends of chromosomes.**
2. **Prevents fusion between exposed ends of chromosomes - chromosomal mutation;**
3. **which can disrupt regulatory control of genes on the adjoined chromosomes**

Points 2 & 3 should be mentioned together

4. **Prevents the activation of DNA repair enzymes which will recognize any exposed chromosomal ends as a defect and degrade the chromosomes.**
5. **serves as buffer DNA to protect the organism's genes from being eroded after successive rounds of DNA replication**
6. **so that critical proteins will still be synthesized in the daughter cells despite the shortened chromosomes**

Points 5 – 6 should be discussed together

7. **Cell will also undergo apoptosis after a limited number of cell division / mitosis (~50) / when a critical length of the telomere is reached**
8. **This limits the extent of accumulated mutations and prevents the development of cancer.**

Max 6m

THE END